

PODS 2025 — Conference Report

Group seminar, Algorithms course, UL FRI
(authors to be filled in)

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1 The conference at a glance

The 44th ACM SIGMOD-SIGACT-SIGAI Symposium on Principles of Database Systems (PODS 2025) was held jointly with SIGMOD 2025 in Berlin, Germany, in June 2025. The proceedings appear as Volume 3, Issue 2 of the *Proceedings of the ACM on Management of Data* (PACMMOD), published in May 2025 (<https://dl.acm.org/toc/pacmmod/2025/3/N2>). The volume opens with an editorial by Carmeli, Geerts, Kara and Kimelfeld (Article 87) and contains 30 research papers as Articles 88–117.

This report covers the December 2024 submission cycle of PODS 2025, whose papers populate the issue above. Out of 80 submitted papers, 30 were accepted, an acceptance rate of 37.5%. (For context, across both submission cycles the conference accepted 45 of 127 papers, 35.4%, but the second cycle is published in a separate issue.)

Topics

PODS is the flagship venue for theoretical work on database systems. The 30 papers in this issue cluster around the following themes:

- query languages and their evaluation: conjunctive queries, regular path queries, Datalog, and disjunctive variants;
- information-theoretic and worst-case-optimal join algorithms, including new uses of fast matrix multiplication;
- differential privacy, streaming and sketching algorithms over relational data, including robust variants over noisy inputs;
- randomised data structures and their adversarial robustness, in particular locality-sensitive hashing and sketches;
- graph algorithms, dynamic data structures and complexity theory;
- explainability, provenance and responsibility measures for query answers;
- data integration, schema matching and emerging applications such as quantum optimisation for database problems.

2 Selected paper summaries

We selected three papers spanning very different parts of this landscape: a deep theoretical advance in worst-case-optimal join evaluation that folds fast matrix multiplication into the polymatroid framework; a sharp attack on locality-sensitive hashing that quantifies how fragile its textbook guarantee becomes against an adaptive adversary; and a streaming algorithm for robust distinct-element counting and ℓ_0 -sampling in general metric spaces, with a matching lower bound.

Abo Khamis, Hu, Suciu.

Fast Matrix Multiplication meets the Submodular Width.

PACMMOD Vol. 3, No. 2 (PODS '25), Article 98.

The data complexity of evaluating a Boolean conjunctive query (CQ) is a central question in database theory. The *submodular width* of a query characterises the running time of optimal combinatorial algorithms; the AGM bound, fractional hypertree width, and the PANDA framework live in this combinatorial setting. In parallel, *fast matrix multiplication* (FMM), which multiplies two $n \times n$ matrices in time $O(n^\omega)$ for $\omega < 2.4$, has been used in scattered, ad-hoc results to beat the combinatorial bounds for specific queries such as k -clique counting. A unifying picture of when and how FMM helps was missing.

Abo Khamis, Hu and Suciu close this gap. They introduce the ω -submodular width, a refinement of submodular width parameterised by the matrix-multiplication exponent ω . The new parameter is never larger than the classical submodular width and equals it when $\omega = 3$ (the FMM-free regime). Their main result, Theorem 7.1, states that for every Boolean CQ Q and every rational ω , there is an algorithm that evaluates Q in time $\tilde{O}\left(N^{\omega\text{-subw}(Q)}\right)$, where N is the input size and $\tilde{O}$ hides polylogarithmic factors. This is the first systematic statement that FMM-aided algorithms fit into the same information-theoretic framework that yielded worst-case-optimal joins.

The proof extends the PANDA machinery. Degree configurations of the input relations are modelled as edge-dominated polymatroids; Shannon inequalities (monotonicity, submodularity) bound the sizes of intermediate results; and ω -submodular width is read off from a polymatroid LP. The new ingredients are a formal calculus of *matrix-multiplication expressions* (Definition 4.2) and a corresponding *variable elimination via matrix multiplication* (Definition 4.5), which permits FMM steps as legal moves inside an otherwise standard query plan. A generalised Reset Lemma (Lemma A.7), strengthening the PANDA Reset Lemma to handle ω -Shannon inequalities with proper dominance, is the technical heart of the upper bound.

The paper does more than unify. Table 1 lists concrete query classes — k -cliques, k -cycles, k -pyramids — for which ω -submodular width is strictly smaller than submodular width, yielding new fastest-known algorithms. The framework also extends to disjunctive Datalog rules, suggesting it should be useful well beyond Boolean CQs. Two takeaways stand out: first, fast matrix multiplication — often viewed as an exotic primitive — fits neatly into the polymatroid and Shannon-inequality picture once degree configurations are made central; second, the unification immediately yields new algorithms, not just retrospective explanations of old ones.

Kapralov, Makarov, Sohler.

On the Adversarial Robustness of Locality-Sensitive Hashing in Hamming Space.

PACMMOD Vol. 3, No. 2 (PODS '25), Article 102.

Locality-sensitive hashing (LSH) is a workhorse data structure for approximate nearest-neighbour search in high-dimensional spaces; its variants underlie image and music retrieval, anomaly detection, clustering and many database join primitives. The classical analysis guarantees that, for a query chosen *independently* of the data structure's random hash functions, the probability of a false negative — missing a point within distance r when one exists — is at most a small parameter δ . The guarantee is fragile, however, when the same query stream can react to earlier responses. How fast can an adaptive adversary actually break LSH?

For LSH in Hamming space the answer turns out to be: very fast. Kapralov, Makarov and Sohler prove (Theorem 1.1) that an adversary with only black-box access to the LSH query interface — no knowledge of the hash families or seeds — can drive the structure into a false negative on a chosen target point using $O(\log(cr) \cdot \log(1/\delta))$ adaptive queries, where r is the

LSH distance parameter and c the approximation factor. The contrast with the non-adaptive setting is dramatic: blind random sampling needs $\Omega(1/\delta)$ trials, an exponential gap.

The attack (Algorithm 2) is an *adaptive walk* starting from an isolated target point z in the dataset. The adversary samples a query at distance $r - t$ from z , observes which neighbours LSH returns, and then flips individual coordinates of the query to drift it away from z while keeping collisions inside the hash buckets that LSH actually inspects. Lemmas 4.1, 4.5 and 4.7 control the number of hash functions involved, the expected bucket sizes and the collision probabilities across hash families. A Chernoff-bound analysis ties these together to show that the walk succeeds with high probability after the stated number of steps.

The result fills a long-standing gap in the adversarial-robustness literature. While the streaming community has produced a sequence of attacks and defences for randomised sketches over recent years, an analogous attack on LSH itself — arguably the more widely deployed of the two primitives — had been missing. The paper makes the practical implication explicit: any system that exposes LSH outputs to inputs that later queries can be derived from (online retrieval, recommendation, anomaly-detection feedback loops) cannot lean on the textbook δ guarantee. It closes by discussing robustness-aware variants and verification as natural follow-up directions, framing the attack as an opening shot rather than a final word.

Qin Zhang.

Robust Statistical Analysis on Streaming Data with Near-Duplicates in General Metric Spaces.

PACMOD Vol. 3, No. 2 (PODS '25), Article 111.

Real streaming data — sensor traces, image feeds, repeatedly measured quantum states, free-text records — is rarely clean: near-duplicates are the norm. Two items that should be treated as the same entity are typically only *close* under some metric rather than literally equal, and an upfront cleaning step is usually infeasible at stream rates. The *robust* versions of the two foundational streaming primitives — F_0 (the count of distinct items) and ℓ_0 -sampling (uniform sampling from the distinct set) — collapse near-duplicates into equivalence classes and ask for correct behaviour on the resulting universe. Prior work handled these robust variants only in $O(1)$ -dimensional Euclidean space.

Qin Zhang lifts the picture to general metric spaces, in a single pass and sub-linear space. The paper splits the problem into two regimes. For *well-shaped* inputs, in which equivalence classes are well separated under the metric, two upper bounds are given: a streaming algorithm computing a $(1 \pm \varepsilon)$ approximation of robust F_0 in $O(\min\{(1/\varepsilon)\sqrt{m \log(1/\varepsilon)}, F_0\})$ words (Section 2), and a robust ℓ_0 -sampler in $O(\sqrt{m})$ words (Section 3). For arbitrary inputs, the right notion is robust F_0 up to an F_0 -ambiguity parameter τ that captures items lying in the borderlands between equivalence classes; Section 4 gives a $(1 \pm \varepsilon)$ approximation in $O((1/\varepsilon)\sqrt{m \log m})$ words for that setting. A matching lower bound in Section 5 shows that even a 1.1-factor approximation requires $\Omega(\sqrt{m})$ bits, sharply separating the general-metric setting from the classical case where $O(\log n)$ space suffices.

The algorithmic ingredients are familiar in shape but carefully adapted to non-Euclidean geometry: a grid partition takes the place of the cell partition that worked in low-dimensional Euclidean space; a global “zero-function” threshold makes sampling consistent across the stream; robust counts are read off as minimum-cardinality valid partitions; and Chernoff bounds discipline deviations across the hash families used to identify near-duplicates. It is the grid step that turns logarithmic space into $\sqrt{m}$ space: in a general metric the number of buckets the algorithm has to track grows polynomially in m rather than only logarithmically.

The interest of the paper is twofold. Practically, it provides the first principled streaming primitive for robust F_0 and ℓ_0 -sampling in settings whose geometry is not low-dimensional Euclidean: textual edit distance, tree edit distance, image-feature distance, and trace distance

between quantum states. That last setting is the paper’s most striking motivation — learning a d -dimensional quantum state requires $\Omega(d/\varepsilon^2)$ noisy copies, producing a stream of approximately-but-never-exactly-equal observations that classical distinct-element machinery cannot consume directly. Theoretically, the $\Omega(\sqrt{m})$ lower bound is the headline: moving from low-dimensional Euclidean to general metrics is not a small quantitative slowdown but a qualitative jump in how much memory robust streaming inherently needs.

References (proceedings)

- M. Abo Khamis, X. Hu, D. Suci. *Fast Matrix Multiplication meets the Submodular Width*. PACMMOD Vol. 3, No. 2 (PODS ’25), Article 98.
- M. Kapralov, M. Makarov, C. Sohler. *On the Adversarial Robustness of Locality-Sensitive Hashing in Hamming Space*. PACMMOD Vol. 3, No. 2 (PODS ’25), Article 102.
- Q. Zhang. *Robust Statistical Analysis on Streaming Data with Near-Duplicates in General Metric Spaces*. PACMMOD Vol. 3, No. 2 (PODS ’25), Article 111.

Full proceedings: <https://dl.acm.org/toc/pacmmod/2025/3/N2>.